What is quantum computing, and what is it good for?

UMKC Quantum Computing Workshop
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Luke Miller, UMKC · QSAIC

1 Why bother

2 It is not magic

3 What the machine physically is

4 What it is good for, and what it is not

5 Where we are now

Your first slow program

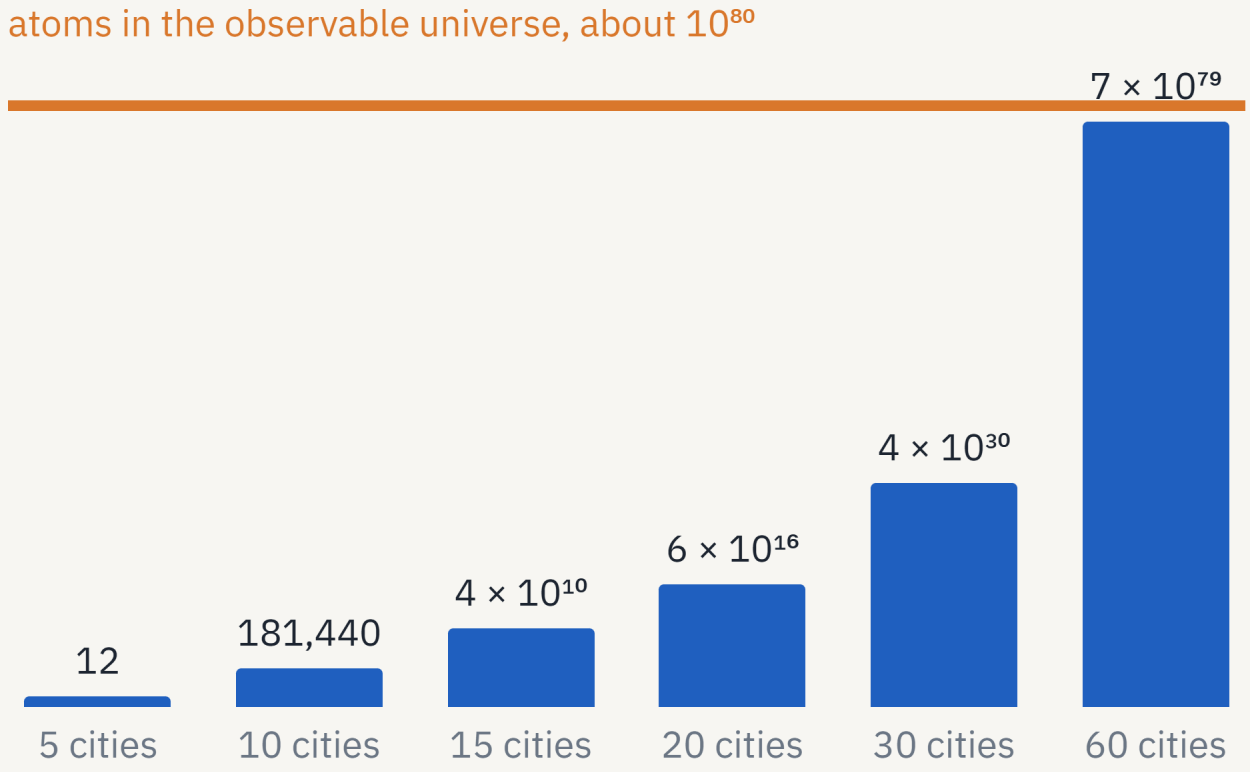
- Everyone here has written a loop that would not finish before the heat death of the universe.
- It was usually trying every combination.

```
for each assignment of 100 bits:
        check whether it solves the problem
number of loop iterations: 2 to the 100th power
```

bits	combinations	at one billion checks per second
30	about 1 billion	1 second
50	about 10^{15}	13 days
70	about 10^{21}	37,000 years
100	about 10^{30}	40 trillion years

The size wall

- A delivery route through 10 cities has 181,440 possible orders.
- A route through 60 cities has about as many orders as there are atoms in the observable universe.



Big compute does not fix exponentials

- A million times more computer buys you seven more cities.
- Moore’s law adds about one bit every two years to a 2^n problem.

machine	route checks per second	largest route solved by brute force in one day
one laptop core	1 billion	17 cities
a GPU cluster	1 trillion	20 cities
an exascale supercomputer	1 quintillion (10^{18})	24 cities

Brute force checks every route. Clever classical algorithms do far better than this, and still hit a wall of their own.

Where quantum enters

- A quantum computer holds a weight on every possible answer at once and steers probability toward the good ones.
- It does not try every answer. It is not magic.

classical: one at a time

00 no

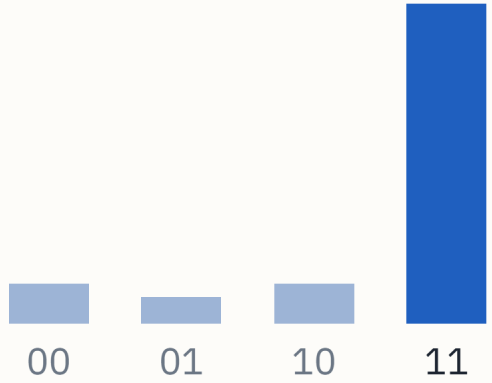
01 no

10 checking

11 not yet

one more bit, twice as long

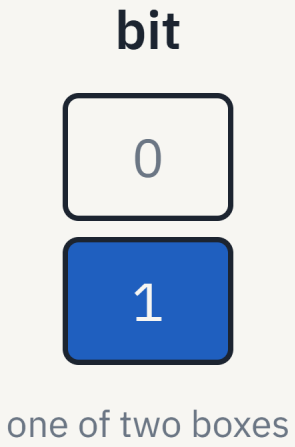
quantum: all weighted at once

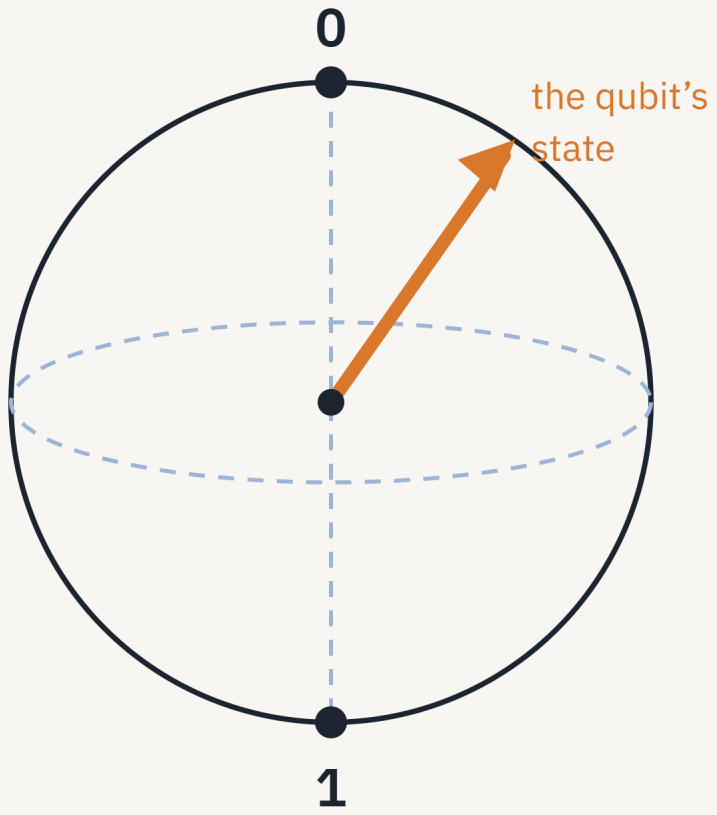


probability of each answer when the algorithm finishes

A bit versus a qubit

- A bit is 0 or 1.
- A qubit is an arrow. It can point at 0, at 1, or anywhere in between.
- Where it points is the qubit's state.





The statevector is a list of numbers

- One qubit is two numbers. Two qubits are four. n qubits are 2^n.
- Each number is called an amplitude, and it belongs to one bitstring.
- Gates change the numbers. Nothing else happens inside.

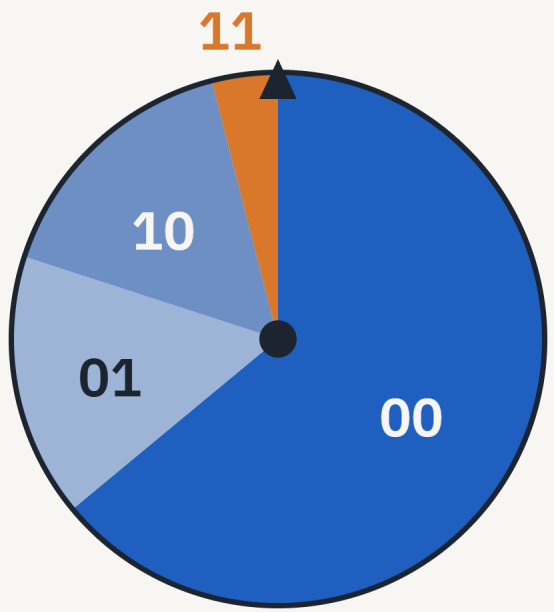
00: 0.8 | 01: 0.4 | 10: 0.4 | 11: 0.2

the whole state of two qubits: four amplitudes, one per bitstring

qubits	numbers in the list
1	2
10	1,024
30	about 1 billion (a workstation can hold it)
50	about 10^{15} (no computer can hold it)
300	about 10^{90} (more than atoms

Measurement picks one bitstring

- Measuring asks the qubits to choose.
- Each bitstring comes out with probability equal to its amplitude squared.
- Afterward the state sits at the answer it gave.

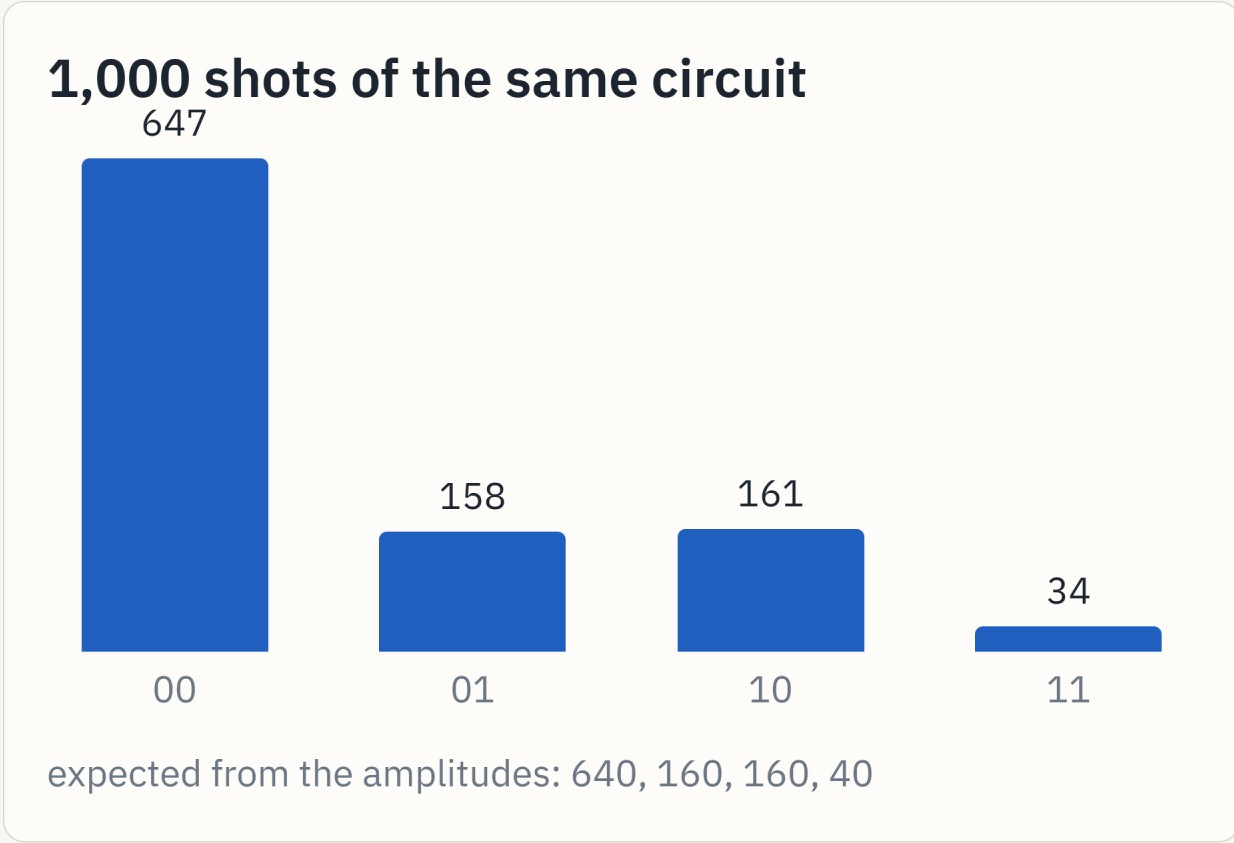


spin it once: one bitstring comes out

bitstring	amplitude	amplitude² = share of spins
00	0.8	0.64 = 64%
01	0.4	0.16 = 16%
10	0.4	0.16 = 16%
11	0.2	0.04 = 4%
total		1.00 = 100%

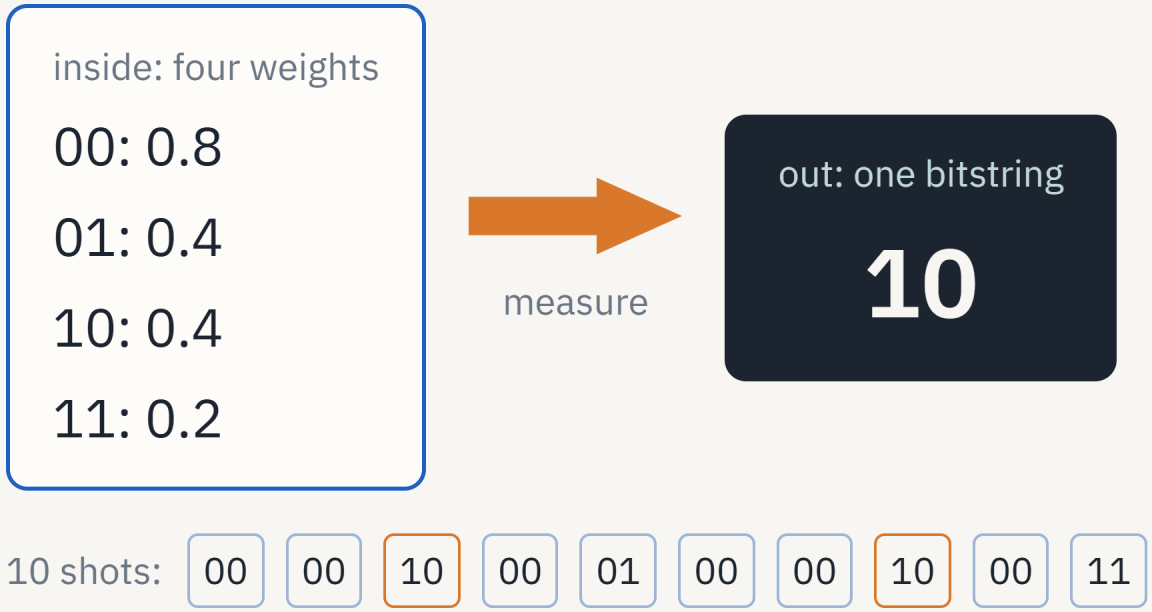
Run it many times

- One measurement is one sample.
- A thousand shots build a histogram.
- The histogram is the answer.



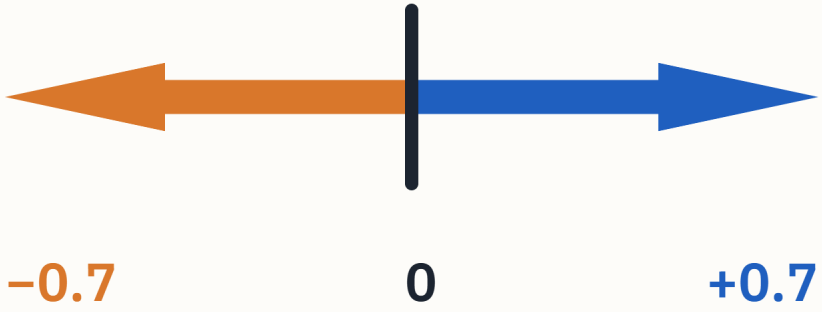
Superposition is weighting, not “both at once”

- The statevector holds a weight for every bitstring.
- Only one bitstring comes out per shot.
- You never see the weights. You see samples.



Amplitudes can be negative

- Probabilities are never negative. Amplitudes can be.
- +0.7 and −0.7 both give probability 0.49.
- The sign only matters when two amplitudes add. That is the next slide.



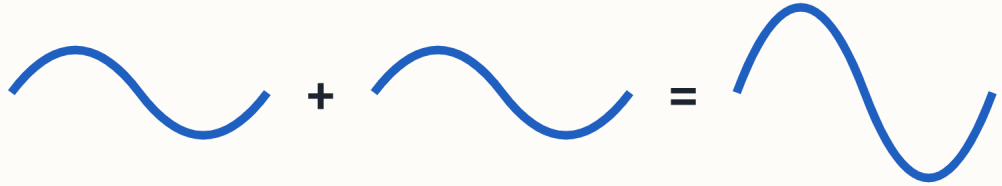
amplitude	amplitude squared
+0.7	0.49
−0.7	0.49

Same probability. Opposite direction. The direction is remembered until you measure.

Interference is the engine

- Two paths to the same bitstring add their amplitudes.
- Same sign reinforces. Opposite sign cancels.
- Noise-cancelling headphones do exactly this with sound.

same sign: reinforce



0.5 + 0.5 = 1.0, so the probability becomes 1.0

opposite sign: cancel

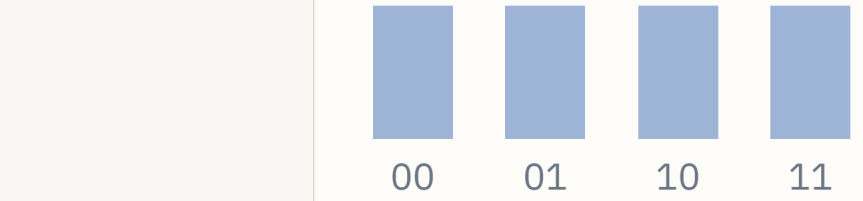


0.5 + (−0.5) = 0, so the probability becomes 0

An algorithm is choreography

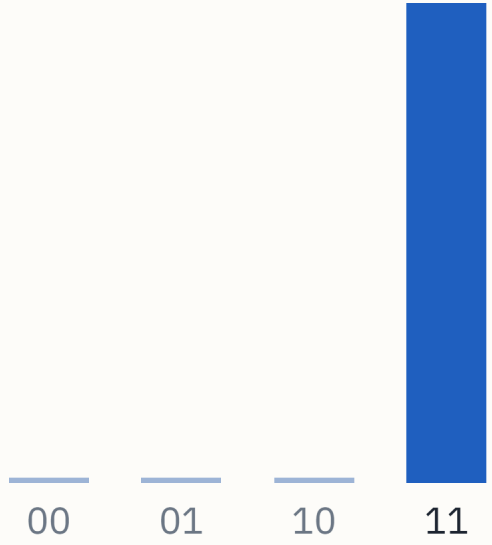
- Gates rotate the arrows.
- A good algorithm routes the wrong answers into cancellation and the right answer into reinforcement.
- Then you measure, and the right answer is what comes out.

before: every answer equal



each at 25%

after: wrong answers canceled



11 at 100%, after one round of Grover’s search

Three gates you will use at 10:30

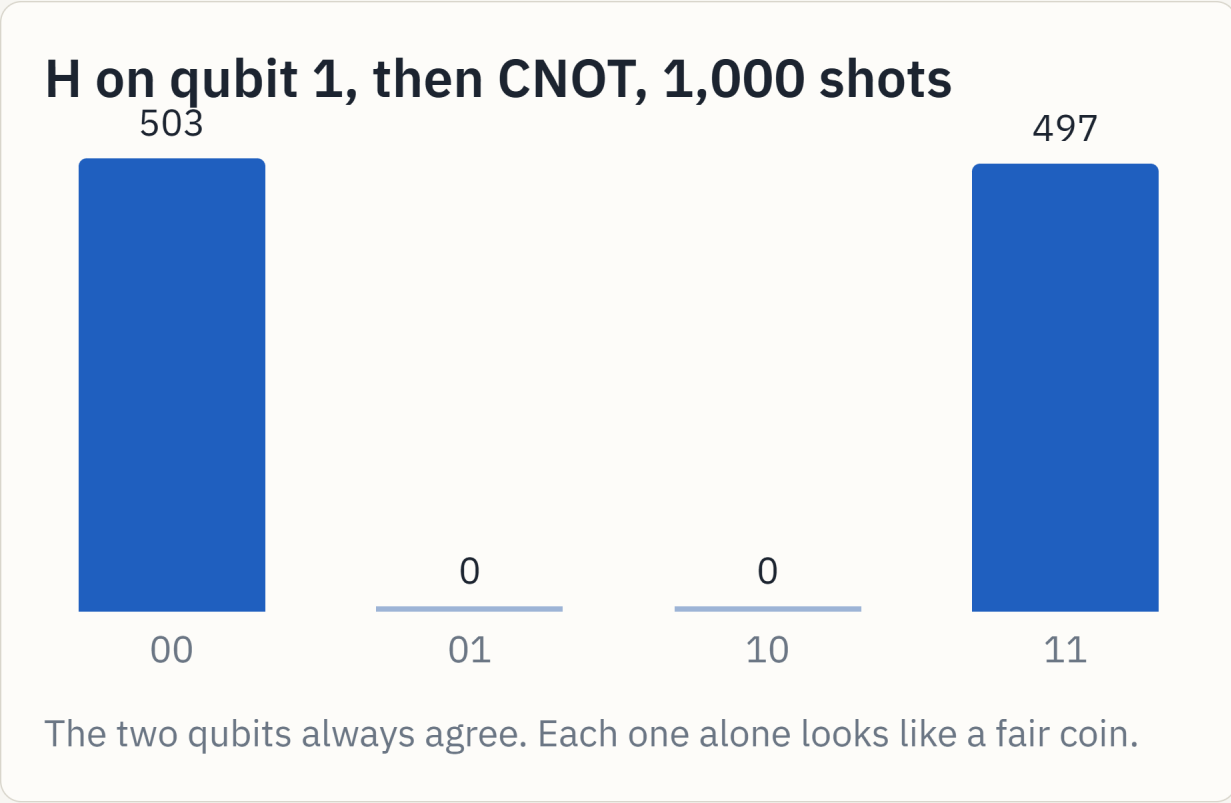
- X flips. H splits evenly. CNOT ties two qubits together.
- Every gate is a rotation of the arrows. None of them looks at the answer.

gate	what it does	before → after
X	flips the arrow	0 → 1, and 1 → 0
H	turns the arrow to halfway	0 → 0.71 on 0 and 0.71 on 1
CNOT	flips the second qubit when the first is 1	00 → 00, and 10 → 11

0.71 is the square root of one half, so each outcome has probability 0.5.

Entanglement

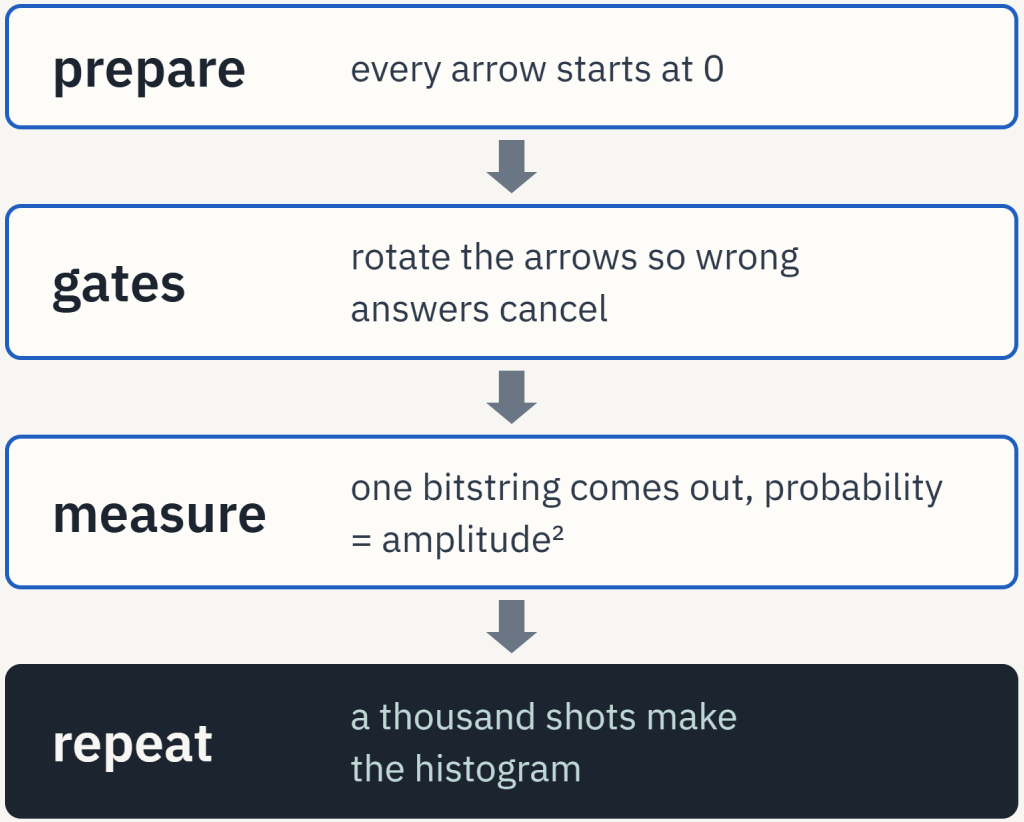
- Two qubits can share one state that cannot be split into two separate arrows.
- Measure one and the other’s answer is fixed.
- It is correlation, not communication. Nothing travels.



The two qubits always agree. Each one alone looks like a fair coin.

Not magic, recap

- Statevector: a list of 2^n numbers.
- Gates: rotate the list.
- Measure: one bitstring, with probability amplitude squared.
- Repeat: the histogram is the answer.



What a quantum computer physically is

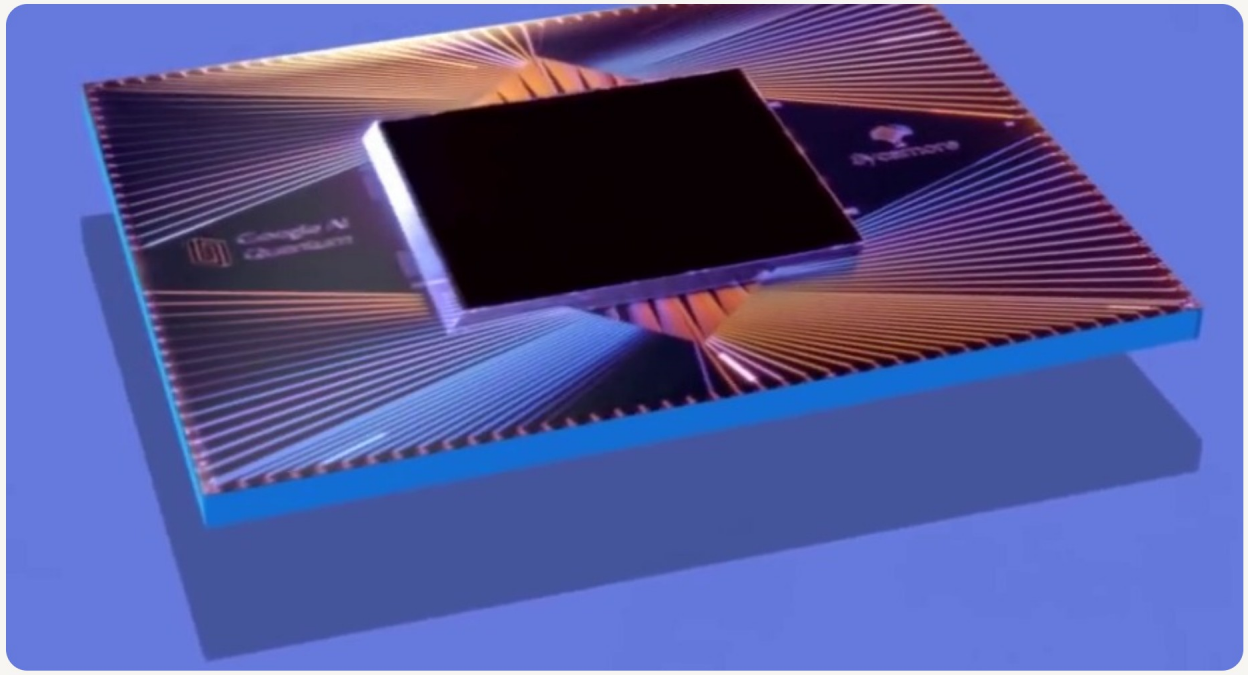
- A qubit is any two-level thing you can control precisely.
- The rest of the machine keeps it cold, quiet, and addressable.
- Most of what you see is refrigerator and wiring.



IBM Quantum System One at Fraunhofer, Ehningen. Photo: IBM Research, CC BY 2.0, via Wikimedia Commons.

Superconducting qubits

- A transmon is a tiny superconducting circuit. Its lowest two energy levels are the 0 and 1.
- Microwave pulses are the gates. Each pulse rotates the arrow.
- IBM, Google, and Rigetti build this kind. It is what you will run on today.



Google Sycamore processor, 53 qubits. Image: Google, CC BY 3.0, via Wikimedia Commons.

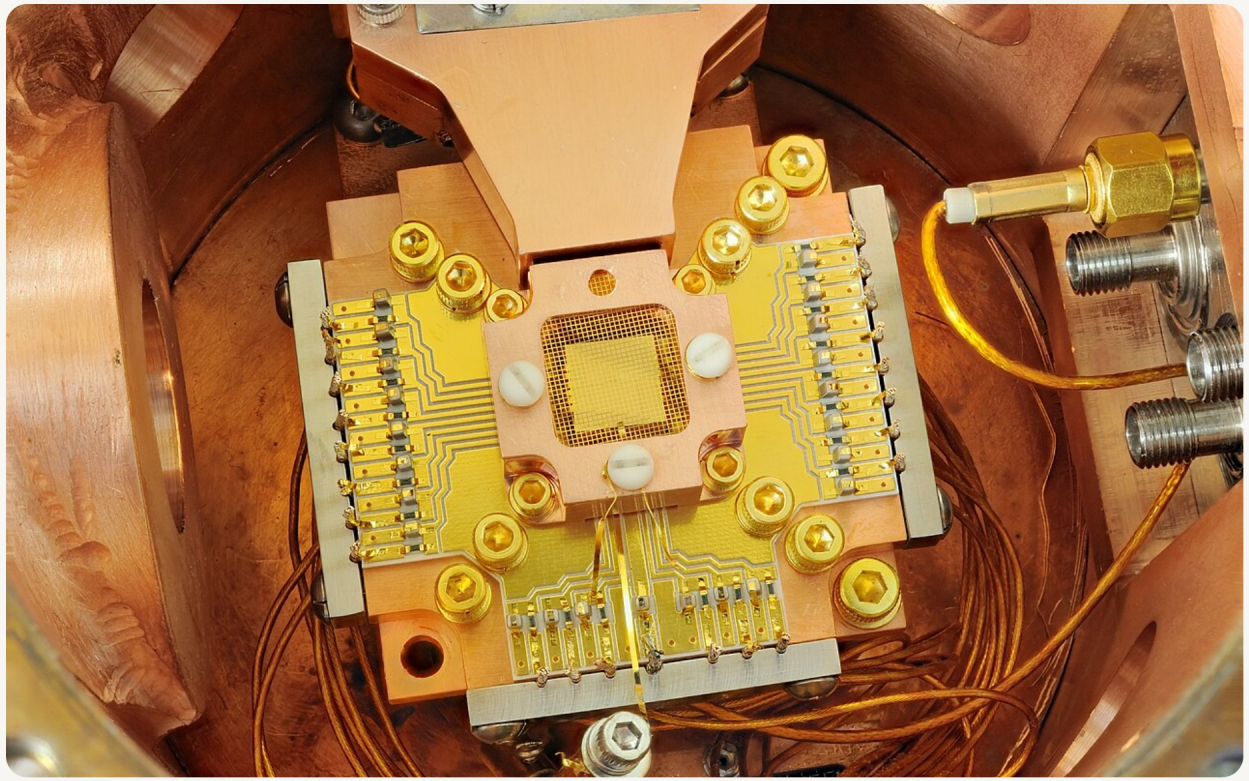
Other ways to build a qubit

- Any two-level system you can isolate and steer will do. Each choice trades speed, connectivity, and error rate differently.

modality	the qubit	kept at	who builds it
superconducting	current in a tiny circuit	0.015 K	IBM, Google, Rigetti
trapped ion	an ion held by electric fields	ions laser-cooled	IonQ, Quantinuum
neutral atom	an atom held by a laser tweezer	atoms laser-cooled	QuEra, Pasqal, Atom Computing
photonic	a particle of light	room temperature	PsiQuantum, Xanadu
spin in silicon	electron spin in a silicon dot	about 1 K	Intel, Diraq
annealer	superconducting loops, optimization only	0.015 K	D-Wave

A trapped-ion chip

- The ions float above the gold electrodes in the center square.
- Lasers do the gates and the readout.
- Same statevector, same gates, same histograms. Different physics underneath.

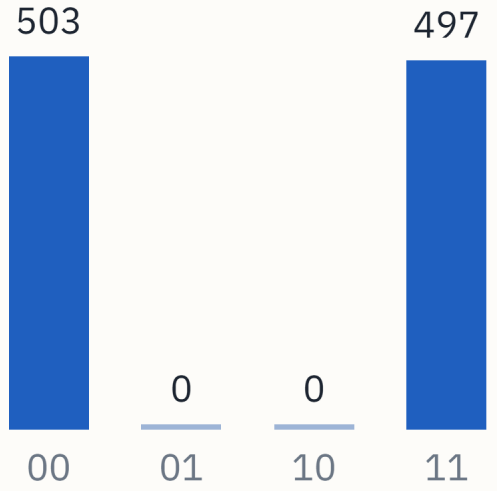


Ion trap at NIST. Photo: National Institute of Standards and Technology, public domain, via Wikimedia Commons.

Noise

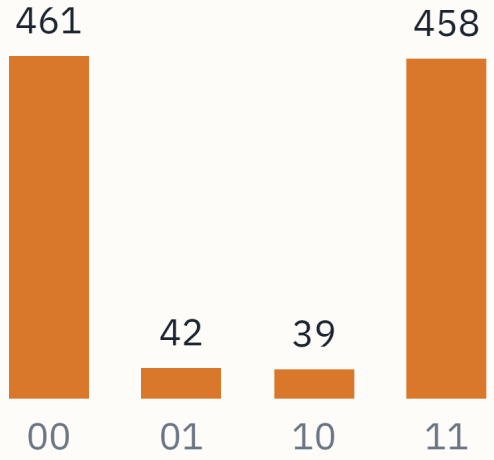
- Qubits forget. Superconducting ones in about a tenth of a millisecond; ions in seconds.
- Every gate has an error rate, roughly 1 in 1,000 for the best two-qubit gates today.
- Deep circuits drown in accumulated error.

simulator



the Bell pair, 1,000 shots, no noise

real hardware



the same circuit on a noisy device: 8% of shots land on bitstrings that should never appear

Good for: chemistry and materials

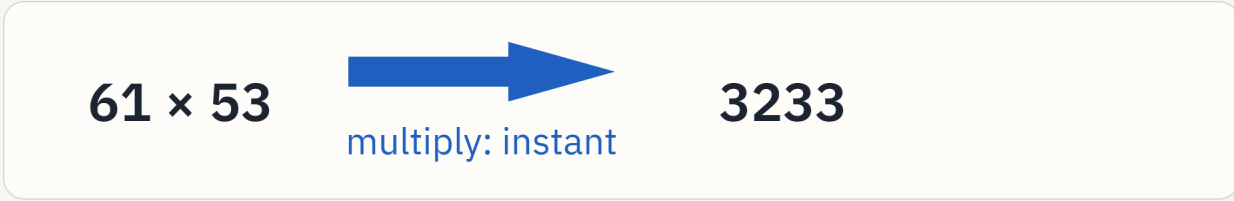
- Molecules are quantum. Their electrons are arrows too.
- Describing them exactly on a classical computer hits the same 2^n wall.
- A quantum computer can hold that description natively.

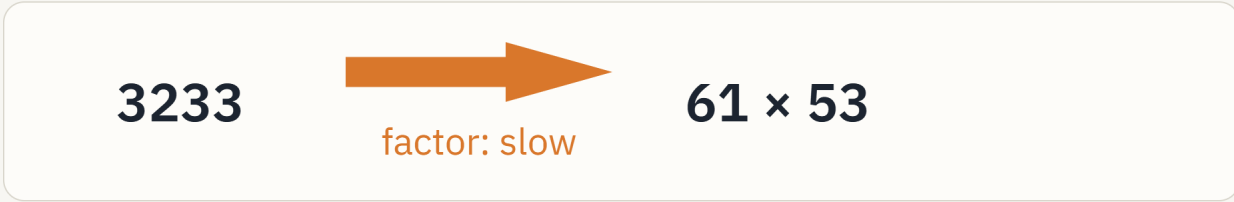
orbitals tracked	numbers in the exact description	roughly
10	1,024	a few atoms
50	about 10^{15}	a small molecule
100	about 10^{30}	a drug-sized molecule
300	about 10^{90}	an enzyme’s active site

Classical chemistry software approximates. The approximations fail exactly where the interesting chemistry is: catalysts, batteries, magnets.

Good for: breaking one kind of lock

- RSA is safe because factoring a big number is slow.
- Shor’s algorithm factors fast, on a machine we do not have yet.
- NIST published post-quantum standards in August 2024. The fix is ahead of the threat.



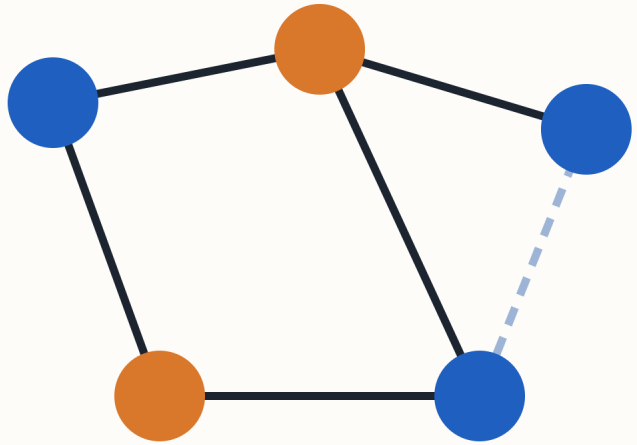


RSA-2048, a 617-digit number	time to factor
best classical method	longer than the age of the universe
Shor, on a fault-tolerant machine	days, with about a million good qubits

Good for: optimization

- Combinatorial problems are the size wall from slide 3.
- QAOA encodes the problem as rotations and steers probability toward good cuts.
- Whether it beats the best classical solvers is an open research question. That is your afternoon project.

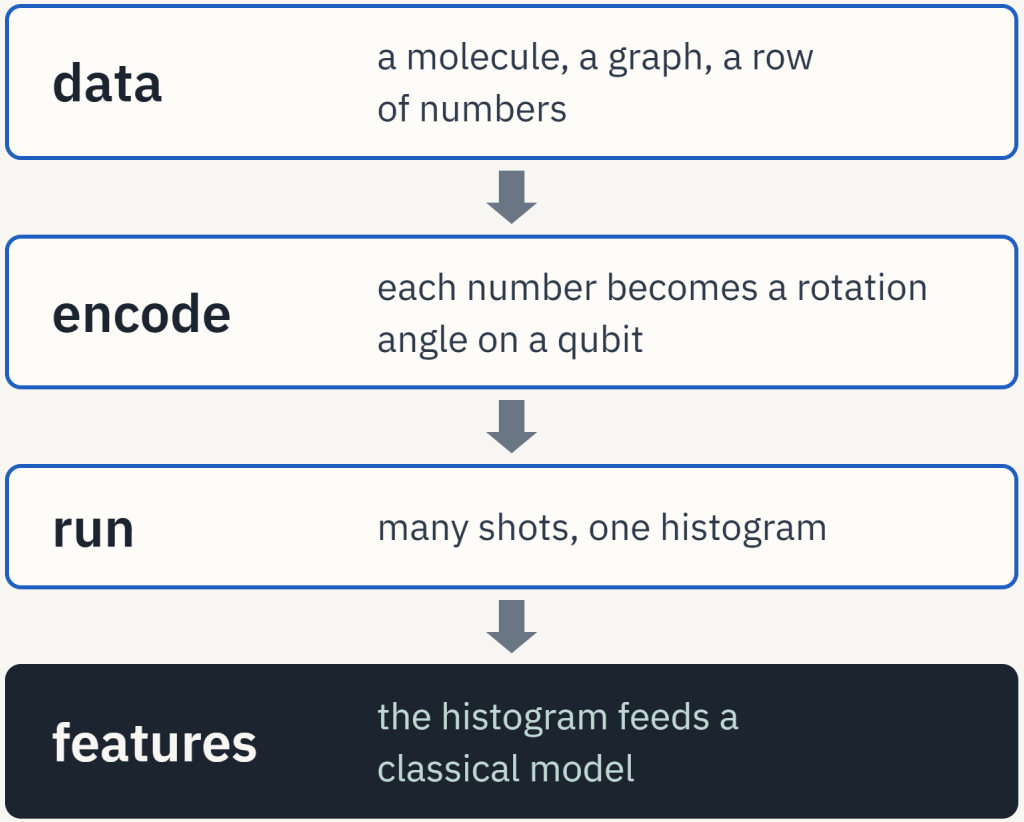
MaxCut: split the nodes so the most edges cross



Blue and orange are the two groups. Five of six edges cross; the dashed edge does not. Each node is one qubit; the measured bitstring is the split.

Good for: machine learning, maybe

- A circuit can turn data into a histogram, and the histogram into features.
- No proven advantage over classical models yet.
- Active research, and a good hackathon topic: the tooling is ready and the questions are open.



What it is not

- Most of what you have heard about quantum computers is one of these.

the claim	what is actually true
It tries every answer at once.	It weights every answer. One answer comes out per shot.
It is faster at everything.	It is faster at a few problem families: simulating quantum systems, factoring, some search and optimization.
It will replace your laptop.	It is a coprocessor you call for one subroutine, the way you call a GPU.
It is science fiction.	Machines with hundreds of qubits run in the cloud today. You will submit a job to one at 10:30.
It is useful in production today.	Not yet for most problems. Today it is for research, learning, and finding out where it will pay off.

The state of the field

- Hundreds to a thousand noisy qubits per machine. Error correction below threshold is demonstrated. Logical qubits are appearing.

year	milestone
2019	Google Sycamore, 53 superconducting qubits: first claim of a task beyond classical reach
2021	IBM Eagle, 127 qubits
2023	IBM Condor, 1,121 qubits, and Heron, 133 qubits with lower error rates
2024	Google Willow: error correction below threshold, a bigger code gives fewer errors
2025	Quantinuum Helios, 98 trapped ions; IBM Nighthawk, 120 qubits on a square grid
2026	[a 2026 milestone you want to cite]

Hard problems are being solved

- Moore’s law has been declared dead every year for fifty years. Expecting progress is not blind optimism; it is the track record.

the objection	where it stands
Qubits forget too fast.	Superconducting coherence times went from nanoseconds in 1999 to hundreds of microseconds today.
Errors can only pile up.	Error correction below threshold was demonstrated in 2024. A bigger code gave fewer errors.
It will never scale past a few qubits.	Thousand-qubit chips exist. Links between chips are being built.
It will never do anything useful.	Chemistry and optimization experiments run on hardware today. The open question is which problems first, not whether.

The day ahead

- Everything after the break is hands on.
- Every circuit you write today is prepare, gates, measure, repeat.
- Bring questions. I will be in the room all afternoon.

10:30	Quantum Hello World and circuit basics
11:45	Introduction to hackathon projects
12:00	Lunch and team formation
1:00	Posters and the elevator pitch
1:30	Project selection, starter notebook
3:15	Make the starter your own
5:00	Done